ProgramBench

Can Language Models Rebuild Programs From Scratch?

Paper: Yang et al., 2026

Guy Tsir Cohen • Agentic AI Systems • November 2026

Agenda

Roadmap



"כדי לשמח את אמא": היזם הישראלי שעשה אקזיט ענק תוך חצי שנה

מאור שלמה חתום על אחד האקזיטים המהירים בתולדות ישראל אחרי שמכר את חברת Base44 ל-Wix ב-80 מיליון דולר • בראיון לתוכנית הטכנולוגיה "טראשטק" הוא מסביר מה דחף להחלטה למכור מהר למרות ההצלחה הגדולה • חושף את המספרים המרשימים: "1 מכל 50 ישראלים השתמש בנו" • ומספר איך המלחמה מול איראן כמעט הרסה הכל

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עקבו אחרינו בגוגל



מאור שלמה, מייסד Base44 | צילום: צילום עצמי

"כדי לשמח את אמא": היזם הישראלי

Thirteen apps.
One craft.

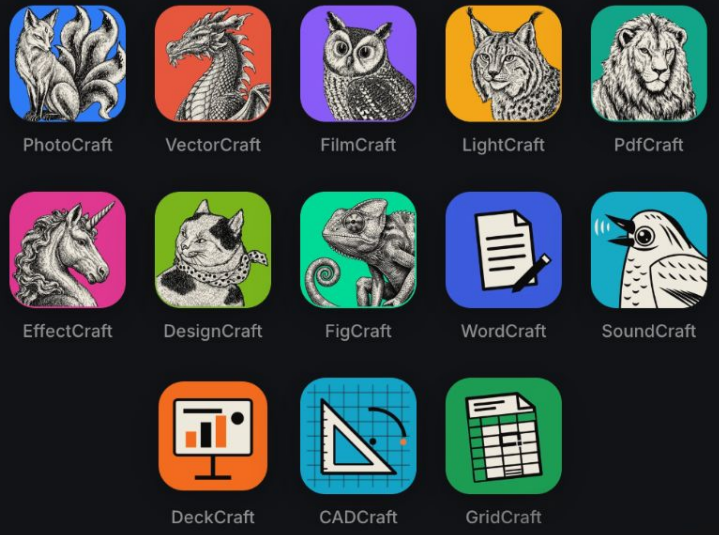
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LINEUP
PRINCIPLES
GET STARTED



מאור שלמה, מייסד Base44 | צילום: צילום עצמי

Context & Motivation

The Problem

0 to 1 Code Generation

Turning ideas written in plain English into fully operational programs is highly popular (Claude Code, Codex, Cursor, etc.)

Benchmark Scope Limit

Existing benchmarks focus on narrow, localized tasks such as fixing a single bug or building a small new feature (e.g. SWE-bench).

Core Research Question

Given an idea, how can we measure the capability of an LM to implement it from scratch?

BACKGROUND

Related Work

SKELETON BENCHMARKS

Commit0 (Zhao et al., 2024): Took 54 Python libraries, stripped out the bodies of all functions and classes, and asked the LLM to fill in the blanks to pass the repository's original unit tests.

DevBench (Li et al., 2024): Provided models with explicit Product Requirement Documents (PRDs) and UML architectural diagrams.

NL2Repo-bench (Ding et al., 2026): Described the expected repository structure in natural language so the agent had to create the files/folders itself, but still dictated the expected internal structure.

SMALL-SCALE PROGRAM RECREATION

Merrill et al., 2026; Chu et al., 2026; Adamczewski et al., 2026

Concurrent works that explored rebuilding programs, but relied on a handful of manually crafted task instances rather than a scalable, automated pipeline.

NEURAL DECOMPILATION (Reverse Engineering)

Tan et al., 2024

Trained models to translate compiled binaries back into the exact original source code.

BACKGROUND

Related Work

BENCHMARK

FEATURED FRAMEWORK

WebGameBench

An end-to-end benchmark environment designed for LLM agents producing interactive web content.

01

SCOPE & CREATION

Creating browser native games from scratch

02

INPUT SPECIFICATION

Uses requirements files fed into an agent

03

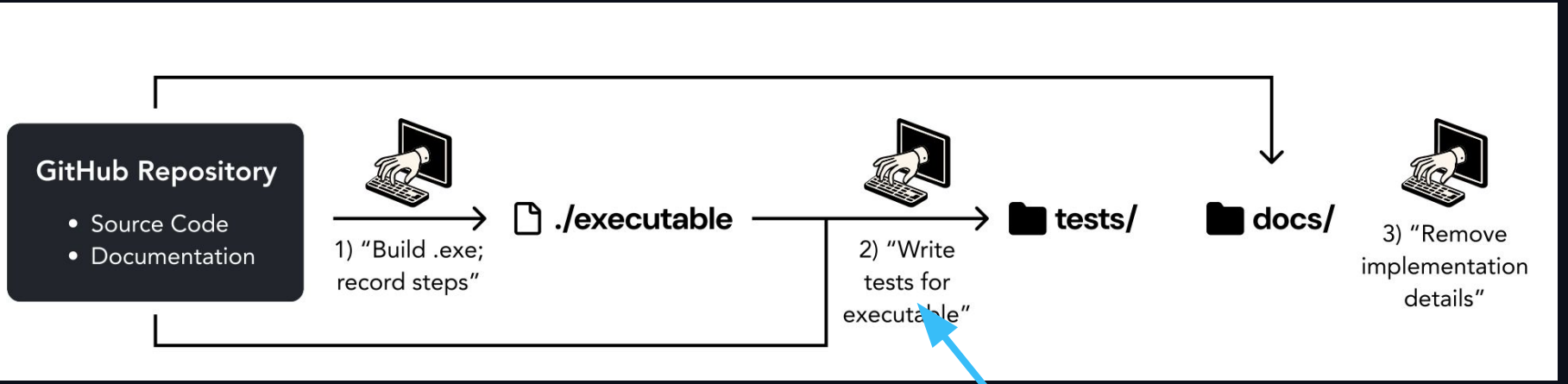
EVALUATION PIPELINE

Uses playwright and Codex to judge the performance of the agent

Back to ProgramBench

Methodology

Task Construction

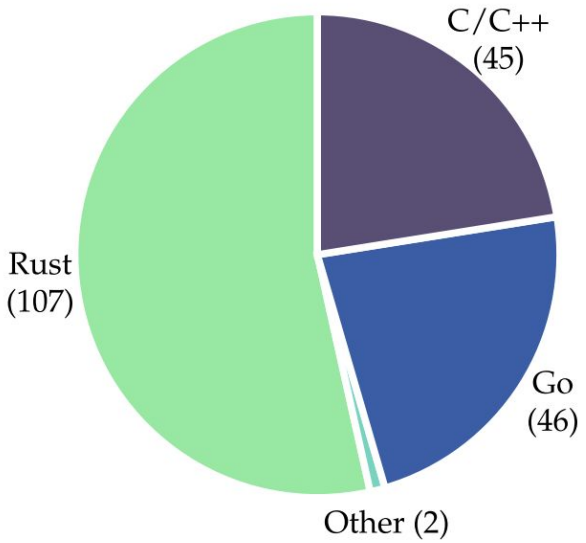


- 200 Tasks
- Based on code written in C,C++, Rust, Go
- Tasks are CLI tools, large-scale popular GitHub repositories like the PHP interpreter

Methodology

Task Construction

Metric	Median	Min	Max
Code lines	8,635	212	2,701,283
Code files	50	1	5,342
Runtime dependencies	10	0	113
Max directory depth	3	0	13
Tests	770	224	14,645
GitHub stars	2,124	202	79,693
Contributors	22	1	422
Commits	646	13	145,991
Repo age (years)	7.9	0.3	17.8



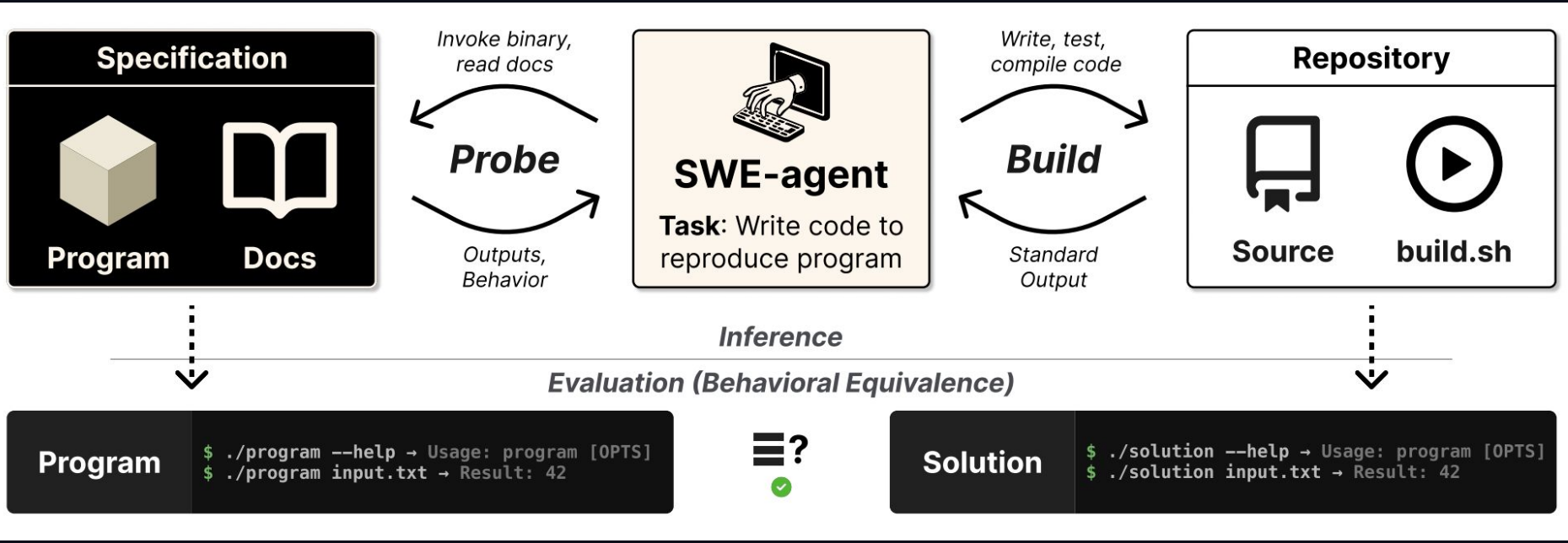
Methodology

Metrics

- Resolved (%) : percentage of task instances where the model's codebase passes all associated test cases
- Almost (%) : percentage of task instances where the model's codebase passes at least 95% of associated test cases
- % Tests Passed
- Cost ($), API calls.

Methodology

The Method



- No internet access
- Docker sandbox
- program only with execute permissions
- Max 1000 steps / 6 hours limit

Discussion

Results - it is hard!

Model	% Res.	% Almost	Calls	$
Claude Opus 4.7	0.0%	3.0%	93	3.81
Claude Opus 4.6	0.0%	2.5%	260	11.38
Claude Sonnet 4.6	0.0%	1.6%	475	27.09
Claude Haiku 4.5	0.0%	0.0%	124	0.80
Gemini 3.1 Pro	0.0%	0.0%	94	1.51
Gemini 3 Flash	0.0%	0.0%	89	0.33
GPT 5.4	0.0%	0.0%	16	0.33
GPT 5.4 mini	0.0%	0.0%	18	0.04
GPT 5 mini	0.0%	0.0%	15	0.03

Discussion

Results - WebGameBench

Table 1: Main results on WebGameBench. Excellent rate and Usable rate use valid scored samples as the denominator; D1–D4 columns report usable/valid counts at each D-level.

Configuration	Coverage	Excellent rate	Usable rate	Usable / valid by D-level			
				D1	D2	D3	D4
opus-4-7 [34]	93.7%	20.2%	76.9%	11/12	17/21	49/65	3/6
opus-4-6 [35]	90.1%	19.0%	73.0%	9/12	17/18	45/63	2/7
gpt-5-5 [37]	99.1%	16.4%	63.6%	8/12	17/20	43/71	2/7
gemini-3.1-pro [38]	96.4%	15.9%	63.6%	10/12	15/20	42/68	1/7
deepseek-v4-flash-thinking [39]	99.1%	13.6%	62.7%	11/12	17/21	40/70	1/7
deepseek-v4-pro-thinking [39]	100.0%	12.6%	62.2%	8/12	18/21	43/71	0/7
kimi-k2.6 [40]	94.6%	16.2%	61.0%	8/12	15/21	41/65	0/7
deepseek-v4-flash-nothinking [39]	97.3%	14.8%	59.3%	11/12	17/21	35/68	1/7
GLM-5.1 [43]	97.3%	12.0%	52.8%	7/12	19/21	31/68	0/7
GLM-5 [42]	99.1%	10.9%	48.2%	8/12	14/21	31/70	0/7
sonnet-4-5 [36]	91.9%	7.8%	47.1%	7/12	15/21	25/63	1/6
deepseek-v4-pro-nothinking [39]	100.0%	16.2%	44.1%	9/12	13/21	26/71	1/7
hy3-xhigh [44]	95.5%	6.6%	41.5%	9/12	14/21	21/67	0/6
kimi-k2.5 [41]	96.4%	8.4%	38.3%	7/11	12/21	22/68	0/7

Discussion

Results (today) - it is getting better FAST

200 tasks · Updated Oct. 22, 2026

Leaderboard | Pareto

RANK	MODEL	AGENT	RESOLVED	ALMOST	COST
1	Claude Opus 5.5 (xhigh)	mini-SWE-agent	18.5%	70.5%	$36.36
2	Claude Fable 5.1 (xhigh)	mini-SWE-agent	9.0%	55.5%	$56.84
3	Muse Wonder Full	mini-SWE-agent	8.5%	60.0%	TBA
4	Muse Wonder (xhigh)	mini-SWE-agent	5.5%	50.5%	$36.53
5	GPT 6 Astra (max)	mini-SWE-agent	5.0%	52.5%	$11.47
6	Claude Opus 5 (xhigh)	mini-SWE-agent	4.5%	37.0%	$50.53
7	Muse Spark 1.3 (max)	mini-SWE-agent	2.5%	25.0%	$6.46
8	Muse Spark 1.3 (xhigh)	mini-SWE-agent	1.0%	16.5%	$2.02
9	GPT-5.6 Sol (xhigh)	mini-SWE-agent	1.0%	15.5%	$6.08
10	GPT 5.5 (xhigh)	mini-SWE-agent	0.5%	13.5%	$8.85

Load more

Discussion

Results - WebGameBench (today?)

Game	Opus 4.6	Opus 5.5
Pair Garden	EXCELLENT	EXCELLENT
Firefly Catch	EXCELLENT	EXCELLENT
Orbit Defense	EXCELLENT	EXCELLENT
Pebble Golf	USABLE	USABLE
Seed Rows	EXCELLENT	EXCELLENT
Lantern Hop	UNUSABLE	EXCELLENT
Hedge Defense	EXCELLENT	EXCELLENT

Context & Motivation

Sub Research Questions

01

LANGUAGE PREFERENCE & CONSTRAINTS

Which languages do models prefer, and what happens when we force a different language?

02

INTERNAL CODEBASE ARCHITECTURE

Do models design clean, modular repositories, or write monolithic spaghetti code?

03

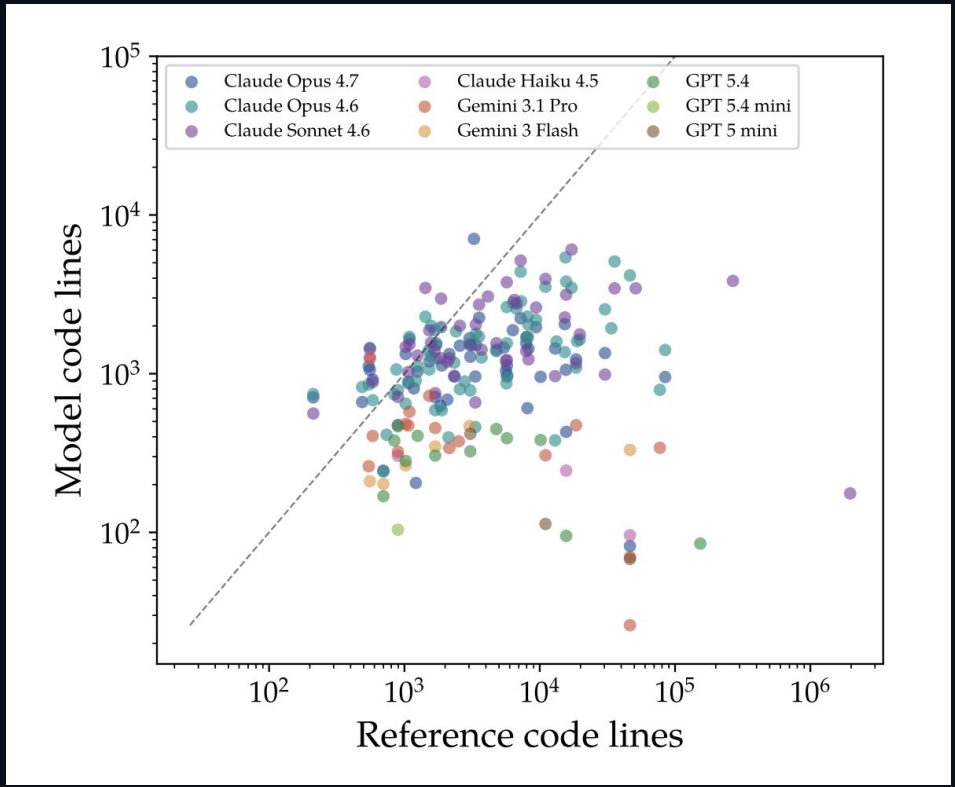
AGENT WORKFLOW & BUDGET ALLOCATION

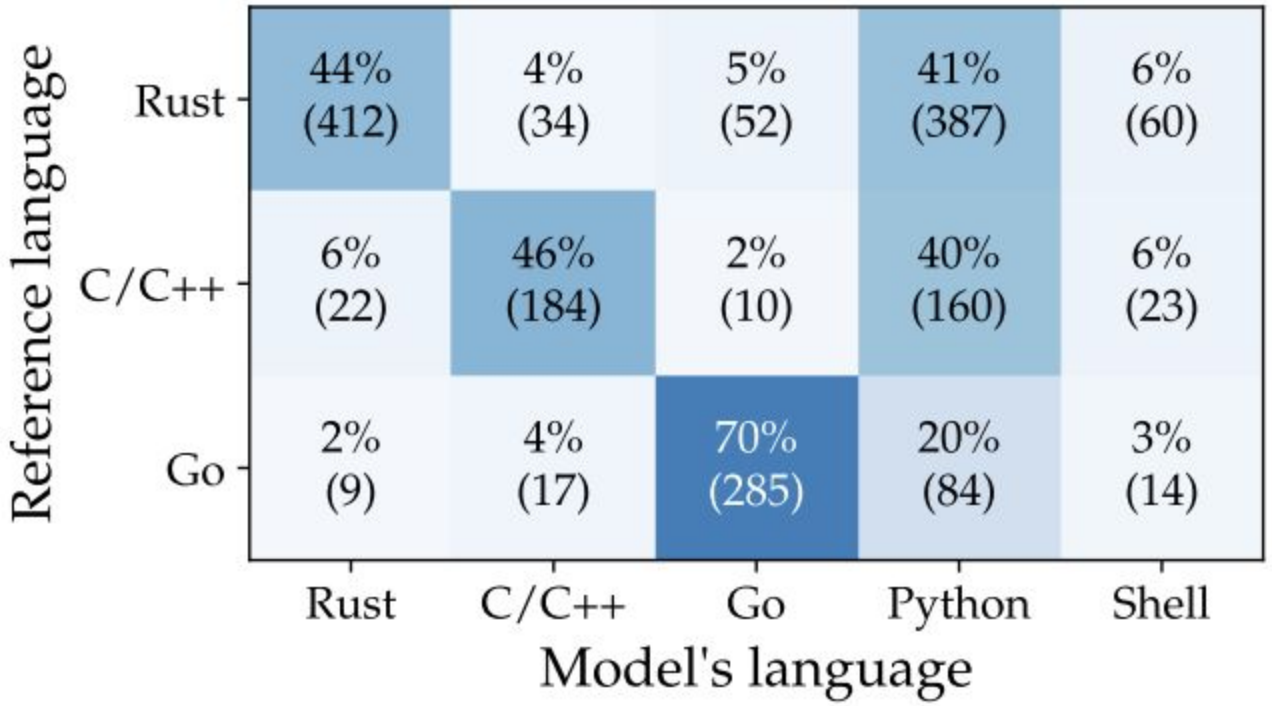
Do agents build and test iteratively like humans, or generate code in a single shot?

Discussion

Further Results

- Models write:
  - Less code
  - Less functions
  - Less files
- Models choose the original language half the time
- GPT uses much less iterations for generation




Reference language
Rust
C/C++
Go
44% (412)
4% (34)
5% (52)
41% (387)
6% (60)
6% (22)
46% (184)
2% (10)
40% (160)
6% (23)
2% (9)
4% (17)
70% (285)
20% (84)
3% (14)
Rust
C/C++
Go
Python
Shell
Model's language

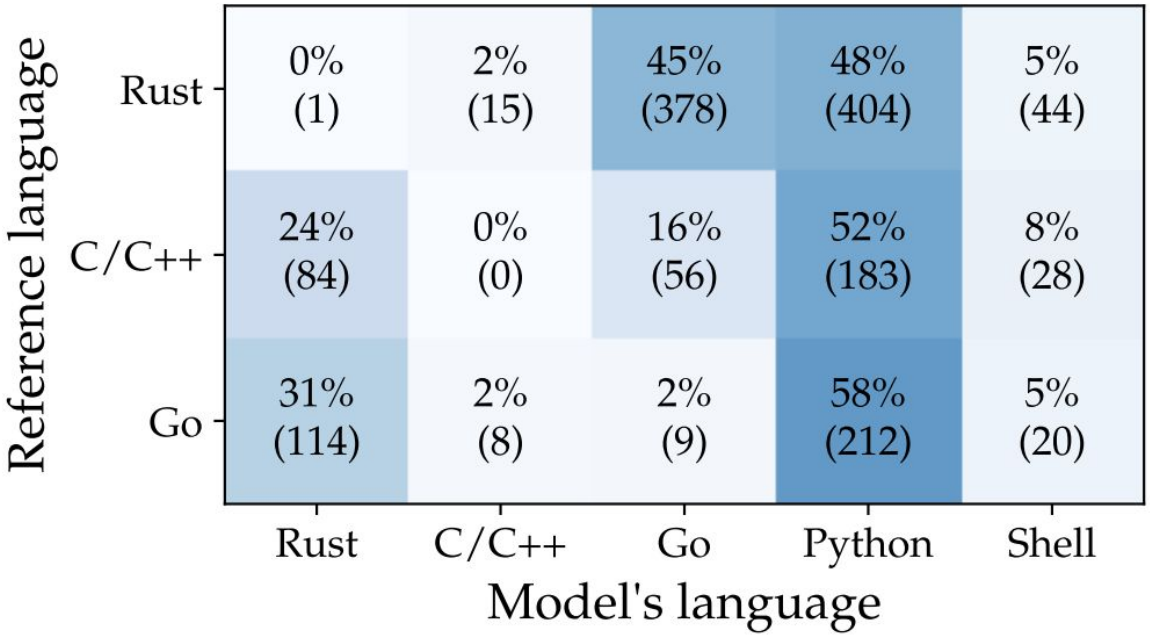


Figure 27 Language confusion matrix under the different-language constraint. Compared to the default setting (Figure 8), the diagonal is nearly empty and Python becomes the dominant choice across all reference languages.

	Opus 4.7		Sonnet 4.6		Gemini 3.1 Pro		GPT 5.4	
	Ref	Model	Ref	Model	Ref	Model	Ref	Model
Function count	133	39 (0.29×)	182	44 (0.24×)	63	10 (0.16×)	88	9 (0.10×)
Avg. length (lines)	25	29 (1.16×)	24	35 (1.46×)	26	42 (1.62×)	24	26 (1.08×)

Discussion

Further Results

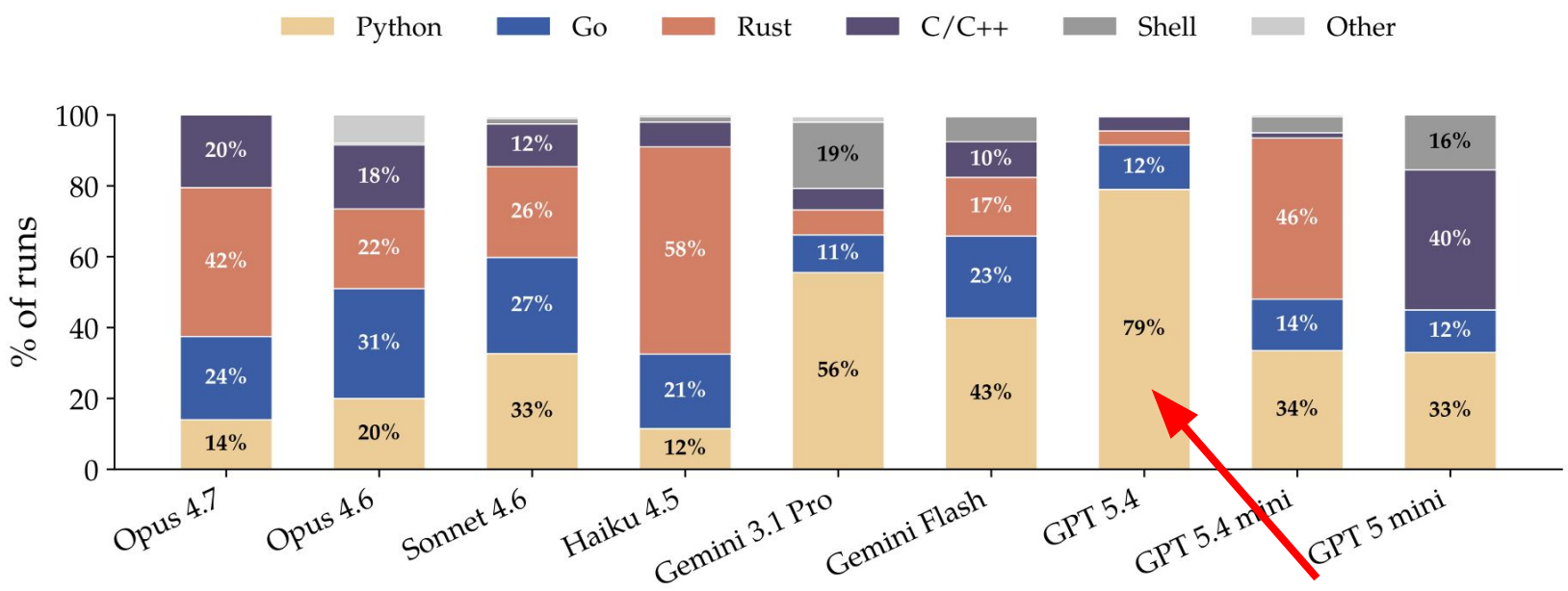


Figure 28 Implementation language chosen by each model, as a percentage of runs. Models are free to choose any language; preferences vary significantly across models.

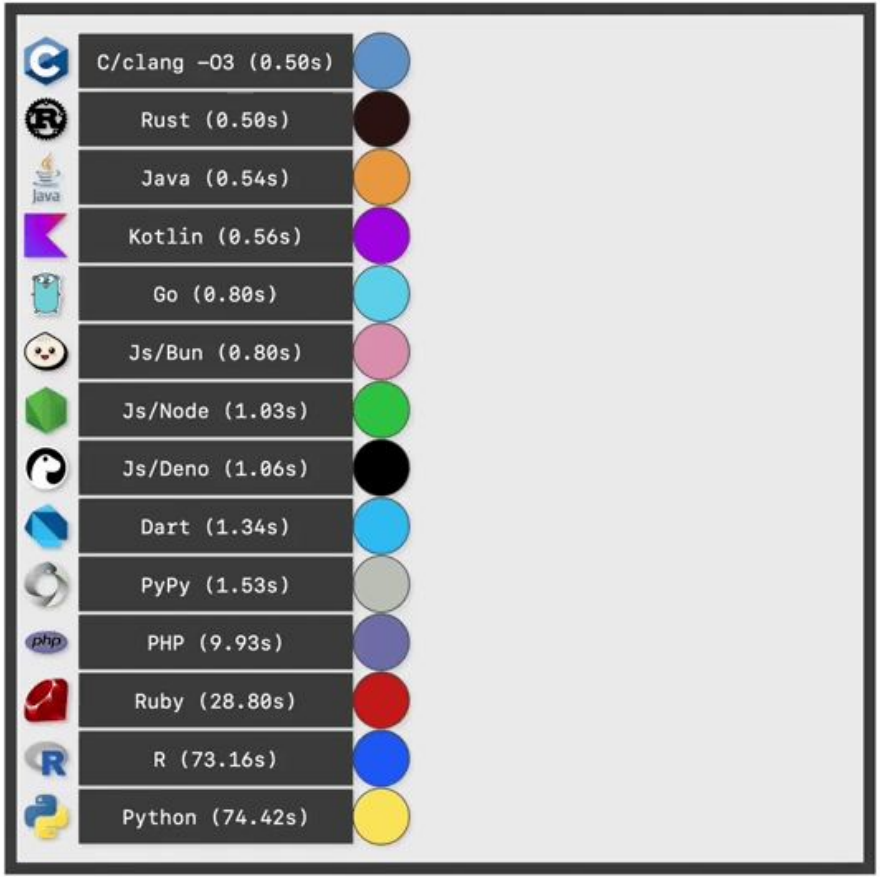
Discussion

Functions f, g are considered equivalent if for every input they return the same output

Discussion

Programs f, g are considered equivalent if for every input they return the same output ?

1 Billion nested loop iterations



Discussion

Programs Equivalence

Runtime

Evaluation focuses purely on input/output matching, masking severe performance disparities.

Program *f* (Rust): 50ms

Program *g* (Python): 45s

Both pass 100% of functional assertions unless a hard timeout is triggered.

Resource & Memory Safety

Memory leaks and critical safety vulnerabilities are completely invisible to standard benchmarks.

- Unfreed memory allocations leaking gigabytes of RAM.
- Unprotected buffer-overflow vulnerabilities left open.
- Receives exact same score as fully memory-safe code.

Architectural Technical Debt

Generated implementations favor monolithic bloat over maintainable software design.

- Thousands of lines crammed into monolithic files.
- Functions are **1.62× longer** than human code.
- Unmaintainable repos fail as viable Day-2 solutions.

Background Tasks & Side Effects

Crucial execution mechanics like caching and background processes are ignored by black-box output checks.

- Omitted caching mechanisms degrading real-world performance.
- Unintended background threads or tasks running silently.
- Side-effects missed by observing input/output matching alone.

Discussion

The Good

- Necessary but insufficient condition for benchmarking models ability to build programs from scratch
- Good insight to language models internals (cheating, language choice, iterations loops
- Hard benchmark for rapidly evolving AI
- More tasks can be added easily

Discussion

Future Work

01 / Metrics Expansion

Adding Execution Speed & Resource Metrics

02 / Structural Analysis

Monolithic vs. Modular Code Architecture

03 / Technical Debt

SWE-Agent Extension Capabilities

Testing the capability of a SWE-agent to extend a project created by itself vs. the original codebase.

04 / Multi-Agent Collaboration

Multi-Agent SWE Benchmarking

Combining multiple SWE-agents on this benchmark to evaluate if collaborative execution yields improved results.